Apr 7 12:58

rvu@rvu-OptiPlex-SFF-7020: ~

GNU nano 7.2 oslab6.sh

```
#!/bin/bash

echo "Lab6: Directory Management"

while true
do
echo ""
echo "1. Create Directory"
echo "2. List Directories"
echo "3. Delete Directory"
echo "4. Exit"

echo "Enter Your choice:"
read choice

case $choice in

1)
   echo "Enter Directory name to create:"
   read dirname
   mkdir "$dirname"
   if [ $? -eq 0 ]; then
      echo "Directory '$dirname' created successfully."
   else
      echo "Error creating directory."
   fi
   ;;
2)
   echo "Directories in current location:"
   ls -d */
   ;;
3)
   echo "Enter Directory name to delete:"
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

Apr 7 12:58

rvu@rvu-OptiPlex-SFF-7020: ~

GNU nano 7.2 oslab6.sh

```
   echo "Enter Directory name to create:"
   read dirname
   mkdir "$dirname"
   if [ $? -eq 0 ]; then
      echo "Directory '$dirname' created successfully."
   else
      echo "Error creating directory."
   fi
   ;;
2)
   echo "Directories in current location:"
   ls -d */
   ;;
3)
   echo "Enter Directory name to delete:"
   read dirname
   rmdir "$dirname"
   if [ $? -eq 0 ]; then
      echo "Directory '$dirname' deleted successfully."
   else
      echo "Error deleting directory (may not be empty)."
   fi
   ;;
4)
   echo "Exiting Program..."
   break
   ;;
*)
   echo "Invalid choice. try again!"
   ;;
  esac
done
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

Apr 7 12:49

rvu@rvu-OptiPlex-SFF-7020: ~

```
^C
(base) rvu@rvu-OptiPlex-SFF-7020:~$ nano oslab6.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~$ chmod +x oslab6.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~$ ./oslab6.sh
Lab6: Directory Management

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter Your choice:
1
Enter Directory name to create:
chandu
Directory 'chandu' created successfully.

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter Your choice:
2
Directories in current location:
324/        b/          Chandana/   documents/  greeshma/   prog/       records/    RVUN25CSE135/  Section/    snap/         system/     Videos/
aditya/     c/          chandu/     Documents/  HANI/       prog1/      RV303/      RVUN25CSE313/  sectionA/   study/        Templates/
admin_lab/  C/          chanduu/    Downloads/  Music/      project/    rvu/        RVUN303/       Sectionc/   Study/        test/
anaconda3/  chandana/   Desktop/    ds_0093/    Pictures/   Public/     RVU/        section/       shrihan/    support_lab/  USN/

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter Your choice:
3
Enter Directory name to delete:
chandu
Directory 'chandu' deleted successfully.
```

```
1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter Your choice:
3
Enter Directory name to delete:
chandu
Directory 'chandu' deleted successfully.

1. Create Directory
2. List Directories
3. Delete Directory
4. Exit
Enter Your choice:
4
Exiting Program...
```

Apr 7 12:57

rvu@rvu-OptiPlex-SFF-7020: ~

GNU nano 7.2 oslab7.sh

```
#!/bin/bash

echo "LAb7: Memory Usage Monitor"
echo ""
echo "Current Memory Status (using free command):"
free -h

echo ""
echo "Detailed Memory Info (using top command):"
top -b -n 1 | grep "Mib Mem"

echo ""
echo "Available Memory:"
free -h | grep "Mem" | awk '{print "Available: " $7}'
```

[Read 14 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

Apr 7 12:51

rvu@rvu-OptiPlex-SFF-7020: ~

```
(base) rvu@rvu-OptiPlex-SFF-7020:~$ ./oslab7.sh
LAb7: Memory Usage Monitor

Current Memory Status (using free command):
               total        used        free      shared  buff/cache   available
Mem:            15Gi       1.5Gi        12Gi       218Mi       1.4Gi        13Gi
Swap:          4.0Gi          0B       4.0Gi

Detailed Memory Info (using top command):

Available Memory:
Available: 13Gi
```

Apr 7 12:57

rvu@rvu-OptiPlex-SFF-7020: ~

GNU nano 7.2 oslab8.sh

```
#!/bin/bash

echo "Lab8: Environment variables Demo "

#Setting variable
name="Chandana"
course="Operating systems"

echo ""
echo "Local Variables:"
echo "Name: $name"
echo "Course: $course"

#Exporting variables
export name
export course

echo ""
echo "Using printenv command:"
printenv name
printenv course

echo ""
echo "All Environment Variables (using env):"
env
```

[Read 25 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

Apr 7 12:52

rvu@rvu-OptiPlex-SFF-7020: ~

```
(base) rvu@rvu-OptiPlex-SFF-7020:~$ nano oslab8.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~$ chmod +x oslab7.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~$ chmod +x oslab8.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~$ ./oslab8.sh
Lab8: Environment variables Demo

Local Variables:
Name: Chandana
Course: Operating systems

Using printenv command:
Chandana
Operating systems

All Environment Variables (using env):
SHELL=/bin/bash
SESSION_MANAGER=local/rvu-OptiPlex-SFF-7020:@/tmp/.ICE-unix/2338,unix/rvu-OptiPlex-SFF-7020:/tmp/.ICE-unix/2338
QT_ACCESSIBILITY=1
COLORTERM=truecolor
XDG_CONFIG_DIRS=/etc/xdg/xdg-ubuntu:/etc/xdg
XDG_MENU_PREFIX=gnome-
GNOME_DESKTOP_SESSION_ID=this-is-deprecated
CONDA_EXE=/home/rvu/anaconda3/bin/conda
_CE_M=
GNOME_SHELL_SESSION_MODE=ubuntu
SSH_AUTH_SOCK=/run/user/1000/keyring/ssh
MEMORY_PRESSURE_WRITE=c29tZSAyMDAwMDAgMjAwMDAwMAA=
XMODIFIERS=@im=ibus
DESKTOP_SESSION=ubuntu
GTK_MODULES=gail:atk-bridge
PWD=/home/rvu
GSETTINGS_SCHEMA_DIR=/home/rvu/anaconda3/share/glib-2.0/schemas
XDG_SESSION_DESKTOP=ubuntu
LOGNAME=rvu
XDG_SESSION_TYPE=wayland
CONDA_PREFIX=/home/rvu/anaconda3
SYSTEMD_EXEC_PID=2401
```

Apr 7 12:53

rvu@rvu-OptiPlex-SFF-7020: ~

```
SYSTEMD_EXEC_PID=2401
GSETTINGS_SCHEMA_DIR_CONDA_BACKUP=
XAUTHORITY=/run/user/1000/.mutter-Xwaylandauth.PH1MN3
HOME=/home/rvu
USERNAME=rvu
IM_CONFIG_PHASE=1
LANG=en_US.UTF-8
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st
=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.t
zo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzst=01;31:*.bz2=01;31:*.bz
=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*
.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=0
1;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:
*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp
4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:
*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;3
6:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xsp
f=00;36:*~=00;90:*#=00;90:*.bak=00;90:*.crdownload=00;90:*.dpkg-dist=00;90:*.dpkg-new=00;90:*.dpkg-old=00;90:*.dpkg-tmp=00;90:*.old=00;90:*.orig=00;90:*
.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
XDG_CURRENT_DESKTOP=ubuntu:GNOME
MEMORY_PRESSURE_WATCH=/sys/fs/cgroup/user.slice/user-1000.slice/user@1000.service/session.slice/org.gnome.Shell@wayland.service/memory.pressure
VTE_VERSION=7600
WAYLAND_DISPLAY=wayland-0
CONDA_PROMPT_MODIFIER=(base)
GNOME_TERMINAL_SCREEN=/org/gnome/Terminal/screen/42d55626_f2e2_459a_b911_784c3882452b
GNOME_SETUP_DISPLAY=:1
LESSCLOSE=/usr/bin/lesspipe %s %s
XDG_SESSION_CLASS=user
TERM=xterm-256color
_CE_CONDA=
LESSOPEN=| /usr/bin/lesspipe %s
USER=rvu
GNOME_TERMINAL_SERVICE=:1.101
CONDA_SHLVL=1
DISPLAY=:0
SHLVL=2
GSM_SKIP_SSH_AGENT_WORKAROUND=true
```

```
GSM_SKIP_SSH_AGENT_WORKAROUND=true
QT_IM_MODULE=ibus
CONDA_PYTHON_EXE=/home/rvu/anaconda3/bin/python
XDG_RUNTIME_DIR=/run/user/1000
CONDA_DEFAULT_ENV=base
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com
XDG_DATA_DIRS=/usr/share/ubuntu:/usr/share/gnome:/usr/local/share/:/usr/share/:/var/lib/snapd/desktop
PATH=/home/rvu/anaconda3/bin:/home/rvu/anaconda3/condabin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap
/bin:/snap/bin
GDMSESSION=ubuntu
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
course=Operating systems
name=Chandana
_=/usr/bin/env
```

Apr 7 12:56

rvu@rvu-OptiPlex-SFF-7020: ~

GNU nano 7.2 oslab9.sh

```
#!/bin/bash

echo "Lab 9:I/O Redirection Demo "

# Output redirection (overwrite)
echo "Hello, This is OS lab" > file1.txt
echo "Content written to file1.txt"

# Append output
echo "Appending new line" >> file1.txt
echo "Content appended to file1.txt"

# Input redirection
echo ""
echo "Reading from file1.txt using input redirection:"
cat < file1.txt

# Error redirection
echo ""
echo "Trying to read non-existing file:"
cat file_not_exist.txt 2> error.txt
echo "Error message stored in error.txt"

# Display error file
echo ""
echo "Error file content:"
cat error.txt
```

[Read 27 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

```
(base) rvu@rvu-OptiPlex-SFF-7020:~$ nano oslab9.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~$ chmod +x oslab9.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~$ ./oslab9.sh
Lab 9:I/O Redirection Demo
Content written to file1.txt
Content appended to file1.txt

Reading from file1.txt using input redirection:
Hello, This is OS lab
Appending new line

Trying to read non-existing file:
Error message stored in error.txt

Error file content:
cat: file_not_exist.txt: No such file or directory
```

Apr 7 13:16

rvu@rvu-OptiPlex-SFF-7020: ~

GNU nano 7.2 oslab10.sh

```
#/bin/bash
# Filename: pipe_demo.sh
# Purpose: Demonstrates command piping in unix shell

echo "Command Piping Demo"

#1. Create a sample file
echo -e "apple\nbanana\napple\norange\nbanana\napple" > fruits.txt
echo "Sample file 'fruits.txt' created with content:"
cat fruits.txt
echo

#2. Count total lines using piping
total_lines=$(cat fruits.txt | wc -l)
echo "Total lines in fruit.txt: $tatal_lines"

#3. Count occurences of 'apple'
apple_count=$(cat fruits.txt | grep "apple" | wc -l)
echo "Number of times 'apple' appears: $apple_count"

#4. Sort the file and remove duplicates
echo "Sorted unique fruits:"
cat fruits.txt |sort | uniq
echo

#5. Search for 'banana' in file
echo "Lines containing 'banana':"
cat fruits.txt | grep "banana"

# Cleanup
rm fruits.txt
echo
echo "Demo complete.Sample file removed."
```

[Read 33 lines]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark M-] To Bracket
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy ^Q Where Was

```
(base) rvu@rvu-OptiPlex-SFF-7020:~$ nano oslab10.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~$ chmod +x oslab10.sh
(base) rvu@rvu-OptiPlex-SFF-7020:~$ ./oslab10.sh
Command Piping Demo
Sample file 'fruits.txt' created with content:
apple
banana
apple
orange
banana
apple

Total lines in fruit.txt:
Number of times 'apple' appears: 3
Sorted unique fruits:
apple
banana
orange

Lines containing 'banana':
banana
banana

Demo complete.Sample file removed.
(base) rvu@rvu-OptiPlex-SFF-7020:~$
```